

Wide Area Network (WAN) Building and Routing Project

Complete Static Routing Design and Routing Document

Prepared by: Engineer Mohamed Ashraf

Target Certification: Cisco Certified Network Associate (CCNA)

Software Used: Cisco Packet Tracer

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1. Project Overview

This document covers the complete software and engineering design for an extensive network project based on full static routing. A square-shaped mechanical topology was built consisting of 4 routers (R1 to R4) to connect 4 independent Local Area Networks (LANs). Each LAN has a dedicated full Class C address range with a fixed /24 subnet mask. Additionally, there are 4 dedicated subnet links for the physical paths connecting the routers (Point-to-Point Links).

2. Network Infrastructure

Based on the physical layout of the topology, the address ranges are distributed as follows:

Local Area Networks (LANs)

- The default range for **LAN1** is 192.168.1.0/24.
- The default range for **LAN2** is 192.168.2.0/24.
- The default range for **LAN3** is 192.168.3.0/24.
- The default range for **LAN4** is 192.168.4.0/24.

Inter-Router Links (WAN Links)

- The top horizontal link (R1 to R3) operates on the 192.168.13.0/24 subnet.
- The bottom horizontal link (R2 to R4) operates on the 192.168.24.0/24 subnet.
- The left vertical link (R3 to R4) operates on the 192.168.34.0/24 subnet.
- The right vertical link (R1 to R2) operates on the 192.168.12.0/24 subnet.

3. Complete Network Addressing Table

Interface / Network Name	Main Network Address	Port IP Address	Subnet Mask	Connection Notes
R1 - GigabitEthernet 0/0	192.168.1.0	192.168.1.1	255.255.255.0 (/24)	Default Gateway for LAN1
R1 - GigabitEthernet 0/1	192.168.12.0	192.168.12.1	255.255.255.0 (/24)	Vertical link towards R2
R1 - GigabitEthernet 0/2	192.168.13.0	192.168.13.1	255.255.255.0 (/24)	Horizontal link towards R3
R2 - LAN Gateway	192.168.2.0	192.168.2.1	255.255.255.0 (/24)	Default Gateway for LAN2
R3 - LAN Gateway	192.168.3.0	192.168.3.1	255.255.255.0 (/24)	Default Gateway for LAN3
R4 - LAN Gateway	192.168.4.0	192.168.4.1	255.255.255.0 (/24)	Default Gateway for LAN4

4. Interfaces Verification

The operational status and software state of the ports were confirmed across all routers using the `show ip interface brief` command. The output retrieved from router R1 demonstrates port stability and active working status.

```
Router# show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.1.1 YES manual up up
GigabitEthernet0/1 192.168.12.1 YES manual up up
```

```
GigabitEthernet0/2 192.168.13.1 YES manual up up
Vlan1 unassigned YES unset administratively down down
```

5. Static Routing Table

To achieve comprehensive connectivity, each router was configured with static routes to access remote networks. The routing table output (`show ip route`) for router R1 distinguishes directly connected networks with a 'C' (or 'L' for local) and statically routed networks with an 'S':

- **S** 192.168.2.0/24 [1/0] via 192.168.12.2
- **S** 192.168.3.0/24 [1/0] via 192.168.13.3
- **S** 192.168.4.0/24 [1/0] via 192.168.13.3

6. Connectivity and Path Tracing Outputs

The network underwent comprehensive testing from endpoint devices to ensure packet delivery occurs without loss.

Part 1: Ping Test

- A ping test was executed from PC1 to PC3 on the remote network at IP 192.168.4.70.
- The results show 4 packets sent, 4 packets received, and 0% loss, achieving an average round-trip time of 12ms.

Part 2: Traceroute Test

A path tracing test (`tracert`) was conducted from PC0 to the LAN4 network, specifically targeting IP 192.168.4.70. The traceroute mapped the following actual hop path:

- **Hop 1:** 192.168.1.1 (First hop: Router Gateway)
- **Hop 2:** 192.168.13.3 (Second hop: Via WAN link to Router R3)
- **Hop 3:** 192.168.34.4 (Third hop: Via vertical link to Router R4)
- **Hop 4:** 192.168.4.70 (Final destination: Targeted endpoint device)

Engineering Report Conclusion

The full static routing configuration has been validated with a 100% success rate. Routes are exchanged accurately across all hops, and the network operates smoothly without the occurrence of any routing loops.